



Integration of DEM-based microstructure generation with FEM and RNM simulations for evaluating ionic conductivity in composite cathodes of all-solid-state batteries

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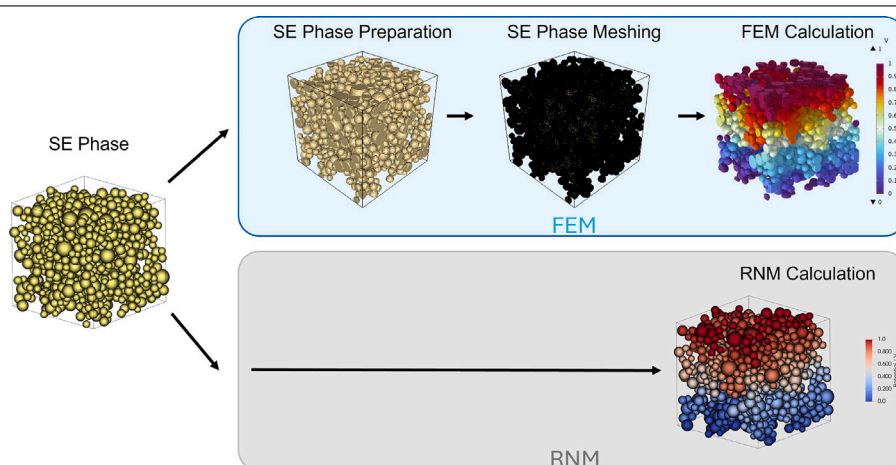
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HIGHLIGHTS

- DEM, FEM, and RNM combined to model SE transport in cathodes.
- FEM and RNM validated using experimental conductivity data.
- RNM is accurate, efficient, and pressure/convergence sensitive.
- SE size and pressure impact percolation and contact quality.
- Network metrics reveal structure–property relationships.

GRAPHICAL ABSTRACT



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ABSTRACT

Cathode microstructure critically governs ionic transport in composite cathodes for all-solid-state batteries (ASSBs). In this study, we present a simulation framework that integrates Discrete Element Method (DEM)-based microstructure generation with Finite Element Method (FEM) and Resistor Network Modeling (RNM) to analyze the effective ionic conductivity $\sigma_{\text{eff,ion}}$ within the solid electrolyte (SE) phase. The framework enables quantitative evaluation of SE network connectivity and transport pathways across realistic 3D microstructures. Both FEM and RNM models were validated against experimental impedance data for composite cathodes with varying active material fractions ($f_{\text{CAM}} = 70\text{--}80\text{ wt.}\%$), demonstrating strong agreement and consistent trends. RNM reproduced FEM results with significantly reduced computational cost, making it suitable for high-throughput parametric studies. Systematic investigations revealed that decreasing the mean SE particle size ($\bar{D}_{\text{SE}}$) enhances $\sigma_{\text{eff,ion}}$ by increasing SE utilization (θ_{SE}) and coordination number ($Z_{\text{SE-SE}}$), despite higher individual contact resistances ($\bar{R}_{\text{SE-SE}}$). Increasing compaction pressure further improves transport

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by densifying particle contacts, though gains diminish beyond 400 MPa. These results establish the DEM-FEM-RNM framework as an efficient and predictive tool for linking processing, microstructure, and ionic transport in ASSB cathodes, providing a foundation for microstructure-informed electrode design and future electrochemical performance modeling.

1. Introduction

The limited energy and power densities of lithium-ion batteries (LIBs) have accelerated the search for advanced storage technologies [1]. All-solid-state batteries (ASSBs) offer a promising alternative, combining higher energy and power densities with improved safety through the use of stable solid electrolytes (SEs) [1]. Among various candidates, sulfide-based SEs have attracted significant interest due to their high ionic conductivities, comparable to those of liquid electrolytes [2]. They also exhibit favorable mechanical properties, including low Young's modulus, ductility, and softness, which promote intimate particle contacts and enable low-temperature processing [3,4].

Even as ASSBs promise higher energy densities and improved safety, achieving reliable performance critically depends on the design of the composite cathode, where ionic and electronic transport bottlenecks, chemomechanical instabilities, and interfacial losses arise from the complex microstructure of cathode active material (CAM) and SE phases [5]. The composite cathode in ASSBs typically comprises a mixture of CAM, SE for ionic transport, and conductive carbon additives for electronic transport such as carbon black [6], carbon nanofibers (CNFs) [7], and carbon nanotubes (CNTs) [8]. Depending on the fabrication process, this powder mixture can be processed into a composite electrode either through a wet slurry-coating method using binders such as ethyl cellulose [9], or via a solvent-free dry-coating method employing polytetrafluoroethylene (PTFE) as a binder [10]. The electrochemical performance of composite cathodes in ASSBs is strongly influenced by their microstructure, where the connectivity of both ionic and electronic phases must be optimized to ensure efficient transport throughout the electrode [11]. In particular, ionic transport through the SE network often represents a critical bottleneck that can severely limit effective utilization and rate capability [12]. Key structural parameters such as the particle size ratio between the CAM and SE phases must be carefully controlled, as a higher CAM-to-SE size ratio has been shown to improve both active material utilization and percolation of the SE phase [13]. Furthermore, applying suitable fabrication pressures during electrode processing can enhance densification and mechanical integrity, although excessively high pressures may induce fracture of CAM particles, disrupting both ionic and electronic conduction pathways [14].

Despite substantial progress, a detailed understanding of ionic and electronic transport limitations within composite cathodes remains challenging; while electronic percolation issues can be mitigated by the addition of conductive carbon additives, achieving sufficient ionic connectivity through the SE phase becomes increasingly difficult at high CAM loadings (f_{CAM}). Indeed, relatively low amounts of carbon additives, typically below 5 wt%, are sufficient to establish an effective electronic percolation network [15], whereas the formation of continuous ionic pathways through the SE remains a critical challenge. Experimental techniques such as X-ray computed tomography (CT) [16–18] and focused ion beam scanning electron microscopy (FIB-SEM) [19, 20] have provided important insights into the three-dimensional microstructure of solid-state electrodes. However, quantifying the direct impact of microstructural features on effective transport properties remains difficult. Therefore, modeling approaches have become increasingly essential to bridge this gap, enabling structure-property relationships to be systematically analyzed and transport limitations to be predicted [13,21–25]. Recently, the Discrete Element Method (DEM) has gained increasing attention for modeling the microstructural formation and behavior of composite cathodes in ASSBs, enabling

studies of mechanical property effects [26], particle packing and arrangement [27], percolation network formation [13], calendaring and processing effects [28], electrical conductivity pathways [29], mechanical degradation during cold-pressing [30], compaction-pressure relationships and ionic conductivity [31], the role of coating structures during particle compaction [32], and input material parameter sensitivity analysis for efficient DEM model calibration [33].

Finite Element Method (FEM) simulations have been widely applied to analyze effective ionic transport properties in battery electrodes. Bielefeld et al. [22] used FEM to compute effective ionic conductivities ($\sigma_{\text{eff,ion}}$) of idealized cathode microstructures, while Clausnitzer et al. [25] extended this approach to structure-resolved simulations with a higher degree of microstructural realism. Although these studies provided important insights, their microstructures were not based on DEM particle-scale generation. In parallel, DEM-based modeling has gained traction for investigating the microstructure of ASSB composites. Sangrós Giménez et al. [29] combined DEM with Resistor Network Modeling (RNM) to study electronic conduction pathways within cathode composites, albeit without explicitly simulating the SE phase. So et al. [30] applied DEM to simulate particle compaction and evaluated transport properties such as tortuosity and relative diffusion coefficients. Schneider et al. [34] combined DEM and FEM to investigate the evolution of ionic conductivity in bulk SE separator structures under pressure, while Yamakov et al. [35] used particle dynamics simulations coupled with RNM to study pressure-dependent ionic conductivity in homogeneous bulk SE materials. However, none of these studies systematically evaluated ($\sigma_{\text{eff,ion}}$) in composite cathodes comprising both CAM and SE phases. Beyond the field of ASSBs, RNM has long been established for transport analysis in granular materials. Early work by Schneider et al. [36] combined DEM with RNM to model the electrochemical performance of sintered solid oxide fuel cell (SOFC) electrodes, while Ott et al. [37] developed micromechanical RNM frameworks for calculating effective conductivities in granular structures. Birkholz et al. [38] notably demonstrated that RNM can provide highly accurate conductivity predictions, with less than 4% error compared to FEM, while offering substantial computational speed advantages. Furthermore, Nisar et al. [39] and Zhang et al. [40] employed DEM and RNM frameworks to study electrical conductivity evolution under mechanical loading, though their focus was on homogeneous rather than heterogeneous battery-like structures. In the context of ASSBs, Ketter et al. [41] recently applied RNM to user-defined cathode composites, providing valuable insights into transport bottlenecks, although without DEM-based microstructural realism. Zhou et al. [42] used DEM to investigate percolation behavior in cathode composites, but did not perform quantitative effective conductivity evaluations.

In this work, we address the critical gap in microstructure-resolved modeling of composite cathodes by integrating DEM, FEM, and RNM into a unified computational framework. Realistic cathode microstructures were generated via DEM using experimentally measured particle size distributions (PSDs) for both the CAM and SE phases, followed by mechanical compaction to capture processing-induced densification. The resulting microstructures were directly employed in FEM and RNM simulations to quantify the effective ionic conductivity $\sigma_{\text{eff,ion}}$ within the percolating SE phase. A convergence analysis was first performed to define the representative volume element (RVE) size for both FEM and RNM calculations. Subsequently, both modeling approaches were validated against experimentally measured $\sigma_{\text{eff,ion}}$ values of baseline cathode composites, demonstrating strong agreement while highlighting the computational efficiency advantage of RNM. The framework was then applied to systematically investigate the effects of SE particle size distribution and fabrication pressure on SE connectivity and

transport. Furthermore, complementary graph-based network analysis was employed to extract microstructural descriptors — such as SE utilization, coordination number, and average contact resistance — to mechanistically link microstructure topology with effective transport. This integrated experimental–computational approach provides a predictive and physically interpretable tool for understanding and optimizing ionic transport in composite cathodes for ASSBs.

2. Experimental methods

All materials were handled exclusively inside an argon-filled glovebox (MBraun GmbH, Germany) with oxygen (O_2) and water (H_2O) concentrations maintained below 0.1 ppm. All materials were utilized as received and stored in sealed containers under an inert atmosphere when handled outside the glovebox, unless mentioned otherwise.

2.1. Materials

The SE used was Li_6PS_5Cl (POSCO, Korea), and the polycrystalline CAM was $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811) (POSCO, Korea). Carbon nanofibers (CNFs) were employed as conductive additives, and polytetrafluoroethylene (PTFE) powder (Chemours, United States) was used as the binder material.

2.2. Particle size analysis

The particle size distribution (PSD) of the SE and CAM materials was measured using a laser diffraction particle size analyzer LA-960V2 (Horiba Ltd., Japan). For sample preparation, a small quantity of each powder was dispersed in butyl acetate and subjected to ultrasonication for 5 minutes to ensure uniform dispersion. All PSD measurements were conducted outside of the glovebox. Given the short measurement time and the dispersion medium used, it is assumed that no significant changes in particle size or morphology occurred during the analysis.

2.3. Material characterization

The CAM and SE samples were analyzed using a scanning electron microscope (SEM) (JEOL JSM-7100F, JEOL, Korea). Sample preparation was carried out inside the glovebox under an Ar atmosphere using an inert sample holder, ensuring no exposure to air. The SEM imaging was conducted at an accelerating voltage of 5 kV.

Phase composition of the SE and CAM was examined by X-ray diffraction (XRD) on a Rigaku powder diffractometer (Rigaku, Japan) equipped with a $Cu K\alpha$ source. The powders were loaded into an airtight sample holder inside the argon glovebox, and diffraction patterns were acquired over a 2θ range of 10° to 90° with an increment of 0.01° .

2.4. Cathode composite preparation

The cathode composites were prepared via a solvent-free, dry-film (DF) method, similar to previous work [33]. All mixing steps were conducted using a planetary ball mill (Pulverisette 7, Fritsch GmbH, Germany) equipped with a 45 mL airtight ZrO_2 jar and 5 mm ZrO_2 balls, maintaining a powder-to-ball ratio of 1:5. Initially, the CAM and SE powders were milled together for 15 min in total (5 min of mixing followed by 5 min of rest) at 300 rpm. Subsequently, the carbon nanofibers (CNFs) were added, and the mixture was milled for an additional 10 min at 200 rpm (5 min cycles of mixing and resting). Finally, the PTFE binder was introduced, and the mixture was milled for another 5 min at 100 rpm. The resulting powder was transferred to a mortar grinder (Pulverisette 2, Fritsch GmbH, Germany) preheated to $100^\circ C$ and processed until a dough-like flake was obtained (approximately 2 min). This flake was rolled into freestanding cathode sheets at $100^\circ C$ to a thickness of approximately 165 μm .

Table 1

Material densities used in this work.

Description	Parameter	Unit	Value	Source
Density (CAM)	ρ_{CAM}	$g\ cm^{-3}$	4.77	[43]
Density (SE)	ρ_{SE}	$g\ cm^{-3}$	1.64	[44]
Density (CNF)	ρ_{CNF}	$g\ cm^{-3}$	1.90	Supplier
Density (PTFE)	ρ_{PTFE}	$g\ cm^{-3}$	2.15	[45]

Table 2

Weight and volume ratios of cathode compositions with varying CAM:SE ratios and fixed additive content (CNF, PTFE).

f_{CAM} (wt.%)	CAM:SE:CNF:PTFE	
	wt.%	vol.%
70	70:27.7:2:0.3	45.3:52.7:1.5:0.5
75	75:22.7:2:0.3	52.2:45.5:1.6:0.6
80	80:17.7:2:0.3	59.5:38.3:1.7:0.6

Three different cathode compositions were prepared, varying the CAM:SE ratio while maintaining constant CNF and PTFE contents. The detailed material densities and the corresponding compositions in terms of weight and volume fractions are summarized in Tables 1 and 2, respectively.

2.5. Bulk SE ionic conductivity measurement

Electrochemical impedance spectroscopy (EIS) was performed to measure the ionic bulk conductivity of the SE. Measurements were conducted using a VMP-300 (Bio-Logic, France) potentiostat. Approximately 100 mg of SE powder was uniformly distributed inside a 10 mm inner diameter polyetheretherketone (PEEK) cylinder between two stainless steel rods, forming an electron-blocking symmetric cell. The SE was densified by uniaxial pressing at 400 MPa for 2 min at room temperature. The assembled cell was placed inside an airtight housing, and a stack pressure was applied during measurement by tightening to a torque of 12.5 Nm (corresponding to approximately 100 MPa). EIS measurements were conducted at room temperature with a voltage amplitude of 10 mV across a frequency range of 1 Hz to 7 MHz.

2.6. Full-blocking cell setup

To measure the effective ionic conductivity of composite cathodes, a full-blocking symmetric cell configuration was employed [19,46–48]. Approximately 100 mg of SE powder was compacted inside a 10 mm inner diameter PEEK cylinder between two stainless-steel rods at 400 MPa for 2 minutes to form a dense SE separator layer. Subsequently, two composite cathode discs (10 mm diameter) were pressed onto both sides of the separator under 400 MPa for 2 minutes. The assembled symmetric cell was mounted into an airtight steel housing, and a stack pressure of approximately 100 MPa was applied using a torque of 12.5 Nm. Electrochemical impedance spectroscopy (EIS) was conducted using a VMP-300 (Bio-Logic, France) potentiostat at room temperature with a voltage amplitude of 10 mV over a frequency range of 1 Hz to 7 MHz. All assembly steps and measurements were performed in an Ar atmosphere at room temperature.

The impedance spectra were fitted using a Z-type transmission-line model (TLM) [19,49] to describe the distributed ionic and electronic transport within the composite cathode. The equivalent circuit topology employed for fitting is illustrated in Fig. 4. The model consists of a repeating ladder network composed of an ionic resistance (R_{ion}), an electronic resistance (R_{elec}), and a constant-phase element (CPE) defined by the admittance $Q(j\omega)^\alpha$, which represents the non-ideal interfacial polarization between both phases. The Z-type configuration corresponds to a fully blocking setup, where both current collectors block ionic transport, leading to a 45° slope at high frequencies and a capacitive tail at low frequencies in the Nyquist representation. A

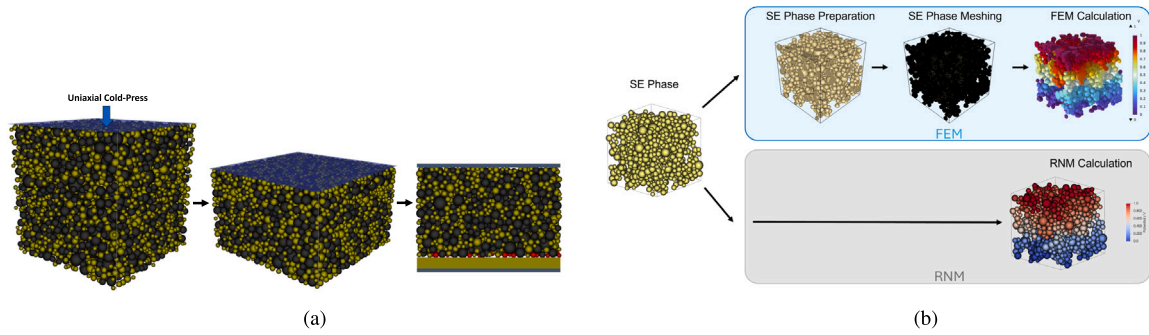


Fig. 1. (a) Schematic of the DEM microstructure generation process, from initial random packing to final compacted electrode structure, demonstrating the CAM particles (black) and SE particles (yellow). After reaching the prescribed pressure, the achieved microstructural data is used for our evaluations, considering the target SE particles (red) which are contact with the SE separator layer (yellow sheet). (b) Workflow comparison between the FEM and RNM modeling approaches. The FEM pipeline (top) involves SE phase extraction, meshing, and electrostatic simulation to compute the potential field and effective ionic conductivity. In contrast, the RNM pipeline (bottom) bypasses geometric meshing and directly solves the resistor network formed by SE particle contacts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

detailed mathematical derivation of the Z-type model and its boundary conditions is provided by Siroma et al. [49].

Parameter estimation was performed using the RELAXIS-3 software. Initial values were assigned based on the order of magnitude observed from preliminary fits. Fit errors were consistently below 2%, and no significant parameter cross-correlation was observed, confirming circuit identifiability. The complete fitting data, including all parameters, cathode layer thickness, electrode area, and fitting errors for each replicate, are provided in Table S1 of the Supplementary Information.

Since the symmetric full-blocking cell contains two identical composite cathode layers separated by the dense SE, the fitted total resistance corresponds to both layers. Therefore, the ionic resistance of a single cathode layer was obtained by dividing the fitted value by two [19]. The resulting effective ionic conductivity, $\sigma_{\text{ion,eff}}$, was calculated according to

$$\sigma_{\text{ion,eff}} = \frac{l}{R_{\text{ion}} A} \quad (1)$$

where l is the thickness of one cathode layer, A is the electrode area, and R_{ion} is the half-cell ionic resistance derived from the fit. The reported $\sigma_{\text{ion,eff}}$ thus represents the average partial ionic conductivity of both composite cathode layers within the symmetric cell.

3. Computational methods

3.1. Discrete element model (DEM)

The DEM simulations in this study were based on our previous work [33] and performed using the open-source software package LIGGGHTS [50,51]. To efficiently model large systems while minimizing computational cost, spherical particles were assumed for both CAM and SE phases. This assumption enables analytical calculation of contact radii and coordination statistics while maintaining numerical robustness during particle compaction. Particle–particle interactions were described using a spring–dashpot model in LIGGGHTS [52], where the elastic term in the normal direction is represented by a Hertzian spring, and the damping term is modeled by an additional spring in the tangential direction [52].

The modeled system included only CAM and SE particles, whereas the conductive CNF and PTFE binder phases — present at a combined content of approximately 2.3 wt.% — were accounted for in the total mass balance but not explicitly included in the DEM domain. Their exclusion is justified by their negligible volumetric contribution and limited influence on the ionic percolation network, which governs the effective transport behavior of the SE phase. The adopted morphology thus captures the dominant structural characteristics relevant to ionic conductivity while maintaining computational efficiency for large-scale

parametric studies. The SE phase was considered as the sole ion-conducting medium in all simulations, consistent with the significantly lower Li-ion diffusivity in the CAM phase [53]. The total system mass m_{tot} was adjusted according to

$$m_{\text{tot}} = \frac{m_{\text{CAM}} + m_{\text{SE}}}{1 - (f_{\text{CNF}} + f_{\text{PTFE}})} = \frac{m_{\text{CAM}} + m_{\text{SE}}}{1 - f_{\text{add}}} \quad (2)$$

where m_{CAM} and m_{SE} are the masses of the CAM and SE phases, respectively, f_{CNF} and f_{PTFE} are the mass fractions of CNFs and PTFE, and $f_{\text{add}} = f_{\text{CNF}} + f_{\text{PTFE}}$ is the total additive mass fraction.

Accordingly, the adjusted mass fractions of CAM and SE in the DEM model were $f_{\text{CAM}}/(1 - f_{\text{add}})$ and $1 - f_{\text{CAM}}/(1 - f_{\text{add}})$, respectively, where f_{CAM} is the CAM mass fraction in the full electrode formulation.

Particles were subjected to gravitational acceleration, and the top simulation wall moved downward at a controlled maximum velocity of $0.2 \mu\text{m}/\mu\text{s}$. A Proportional-Integral-Derivative (PID) controller maintained a constant uniaxial compaction pressure on the system. Once the target pressure was reached, the system was allowed to relax until the total kinetic energy fell below a critical threshold. Time-step sizes were selected to satisfy both the Rayleigh and Hertz stability criteria [54]. The initial simulation domain was defined as a cuboid box, with lateral dimensions of $10\bar{D} \times 10\bar{D}$ and a vertical dimension of $15\bar{D}$, where $\bar{D}$ is the mean particle diameter of the larger particle type [13,33]. Periodic boundary conditions were applied laterally, while the top and bottom boundaries were treated as movable and fixed walls, respectively. The initial particle packing density was selected so that to occupy approximately 75% of the box volume, in order to balance computational efficiency and successful packing. The DEM input scripts were generated using an automated MATLAB® workflow, facilitating the generation of large datasets and enabling parametric optimization studies. An illustration of the DEM microstructure generation process, from initial random packing to the final electrode structure, is shown in Fig. 1(a). To ensure statistical reliability, ten independent DEM realizations were generated for each cathode composition. Mean values and standard deviations were computed and reported for all quantities. All DEM simulations were executed on a Lenovo ThinkSystem SR650 cluster operating under SUSE Linux Enterprise Server.

The material and contact parameters used for the DEM simulations are summarized in Table 3.

3.2. Finite element method (FEM)

The finite element (FEM) approach represents the first part of the integrated computational framework outlined in Fig. 1(b). In this workflow, the 3D microstructures generated by DEM are used as direct input for the FEM simulations. The solid electrolyte (SE) phase is extracted from the composite domain, meshed, and subsequently analyzed to

Table 3
DEM input parameters used in this study.

Parameter	Value	Source
SE-SE friction coefficient (μ_{SE-SE})	0.4	[33]
CAM-CAM friction coefficient ($\mu_{CAM-CAM}$)	0.4	[33]
CAM-SE friction coefficient (μ_{CAM-SE})	0.4	[33]
SE Young's modulus (E_{SE}) [GPa]	22.1	[55]
CAM Young's modulus (E_{CAM}) [GPa]	161.5	[33]
SE Poisson's ratio (ν_{SE})	0.37	[55]
CAM Poisson's ratio (ν_{CAM})	0.30	[43]
Coefficient of restitution (RC)	0.4	[33]

determine its effective ionic transport properties across the electrode thickness.

The FEM simulations were directly based on the 3D cathode microstructures generated by the DEM model. In these structures, each particle type (CAM, SE) was spatially resolved, allowing the solid electrolyte (SE) phase to be digitally isolated as the percolating domain responsible for ionic transport. This extracted SE network was then used as the computational geometry for solving the governing transport equations under an applied potential difference across the electrode thickness. Consequently, the FEM approach establishes a direct link between the DEM-predicted microstructural morphology and the resulting effective ionic conductivity.

To calculate the effective ionic conductivity of the SE particles' phase through the electrode thickness, we first solve the steady-state diffusion transport equation in the microscopic domain Ω_{SE} , governed by

$$\nabla \cdot \mathbf{J} = 0, \quad \mathbf{J} = -\sigma \nabla \phi, \quad \text{in } \Omega_{SE} \quad (3)$$

where ϕ represents the electric potential, and $\mathbf{J}$ is the ionic flux through the SE particle domain. We assign Dirichlet boundary conditions for the upper and lower domain boundaries as $\phi_1 = 1$ V and $\phi_0 = 0$ V, respectively. All remaining boundaries were treated as electrically insulating, enforcing a zero-Neumann condition $-\mathbf{n} \cdot \mathbf{J} = 0$. Solving Eq. (3), the total ionic current flux q over the bottom boundary Γ can be calculated as

$$q = \int_{\Gamma} -\mathbf{n} \cdot \mathbf{J} dS \quad (4)$$

Furthermore, we consider the macroscopic transport Eq. (5) with similar boundary conditions as for Eq. (3), in which the ionic flux $\bar{\mathbf{J}} = -\sigma_{\text{eff,ion}} \nabla \bar{\phi}$ takes place uniformly through the homogeneous domain Ω_{dom} :

$$\nabla \cdot \bar{\mathbf{J}} = 0, \quad \bar{\mathbf{J}} = -\sigma_{\text{eff,ion}} \nabla \bar{\phi}, \quad \text{in } \Omega_{dom} \quad (5)$$

The homogeneous ionic flux $\bar{\mathbf{J}}$ in Eq. (5) can be analytically expressed as $\bar{\mathbf{J}} = \sigma_{\text{eff,ion}} \frac{\Delta \phi}{L}$, where $\Delta \phi = \phi_1 - \phi_0$ and L represents the domain height. Therefore, the total ionic current flux q in this case equals $A_{dom} \bar{\mathbf{J}}$, where A_{dom} is the cross-sectional area of the cuboid macroscopic domain Ω_{dom} , perpendicular to the ionic current flow.

Finally, by comparing the total ionic current flux q in the microscopic and macroscopic cases, the effective ionic conductivity can be calculated as

$$A_{dom} \sigma_{\text{eff,ion}} \frac{\Delta \phi}{L} = \int_{\Gamma} -\mathbf{n} \cdot \mathbf{J} dS = q \Rightarrow \sigma_{\text{eff,ion}} = \frac{q L}{A_{dom} \Delta \phi}. \quad (6)$$

For numerical calculations, the Conductive Media DC interface in COMSOL Multiphysics® was employed, using the GMRES iterative solver, which minimizes the residual within a Krylov subspace to efficiently handle large, sparse linear systems [56].

A schematic overview of the overall FEM-RNM simulation workflow is provided in Fig. 1(b). Here, the FEM branch (described in this section) represents the continuum-scale evaluation of ionic transport, while the parallel RNM branch (discussed in Section 3.3) captures particle-level contact resistances based on the same DEM-derived microstructures.

3.3. Resistor network model (RNM)

To quantify $\sigma_{\text{eff,ion}}$ through the SE phase, a Resistor Network Model (RNM) was developed based on particle-level microstructures obtained from DEM simulations. The RNM was constructed by representing each SE particle as a node in a graph, with interparticle ionic transport modeled via resistive connections between physically contacting particles. Only the percolating SE phase spanning from bottom to top boundaries was considered for conductivity evaluation. The modeling workflow is illustrated schematically in Fig. 2.

The resistances between two contacting nodes were based on contact-limited transport. For a general case where two particles I and J are composed of materials with different conductivities σ^I and σ^J , respectively, the analytical expression for the resistance is given by [38]:

$$R^{I,J} = \frac{1/\sigma^I + 1/\sigma^J}{4r_c^{I,J}}, \quad (7)$$

where $r_c^{I,J}$ is the contact radius between particles I and J . Assuming identical materials for all SE particles ($\sigma^I = \sigma^J = \sigma$), this simplifies to

$$R^{I,J} = \frac{1}{2\sigma r_c^{I,J}}. \quad (8)$$

For interactions between an SE particle J and a flat electrode surface, the resistance is expressed as

$$R^{J,0} = \frac{1}{4\sigma r_c^{J,0}}, \quad (9)$$

where $r_c^{J,0}$ denotes the contact radius formed between the particle and the planar electrode surface.

The contact radii $r_c^{I,J}$ were calculated analytically from the sphere-sphere intersection relation [38]:

$$r_c^{I,J} = \frac{1}{2d^{I,J}} \sqrt{(d^{I,J} + R^I)(d^{I,J} - R^I + R^J)} \sqrt{(-d^{I,J} + R^I)(d^{I,J} + R^I + R^J)}, \quad (10)$$

where $d^{I,J}$ is the center-to-center distance between particles, and R^I, R^J are their radii. For particle-boundary contacts, the contact radius was obtained from the deformation depth δ between the particle and the boundary plane as

$$r_c^{J,0} = \sqrt{r^2 - (r - \delta)^2}. \quad (11)$$

These calculations were implemented in an in-house MATLAB® toolbox to generate the complete symmetric contact-radius matrix.

Once the resistor network was assembled, the node potentials were computed by solving the Kirchhoff current balance at each node i :

$$\sum_j \frac{\phi_i - \phi_j}{R^{i,j}} = 0, \quad (12)$$

where the summation extends over all neighboring nodes j connected to i . Fixed potentials were applied at the top ($\phi_1 = 1$ V) and bottom ($\phi_0 = 0$ V) boundaries, while lateral boundaries were treated as electrically insulating. The resulting sparse linear system was solved following the procedure of Birkholz et al. [38].

After solving for the potentials, the local current through each connection was evaluated as

$$I_{i,j} = \frac{\phi_i - \phi_j}{R^{i,j}}. \quad (13)$$

The total ionic current q was obtained by summing all currents crossing the top boundary:

$$q = \sum_{i \in \Gamma_{\text{top}}} I_{i,j}, \quad (14)$$

where Γ_{top} represents all particle-electrode contacts at the upper boundary. The effective ionic conductivity was then determined from

$$\sigma_{\text{eff,ion}} = \frac{q L}{A_{dom} \Delta \phi}, \quad (15)$$

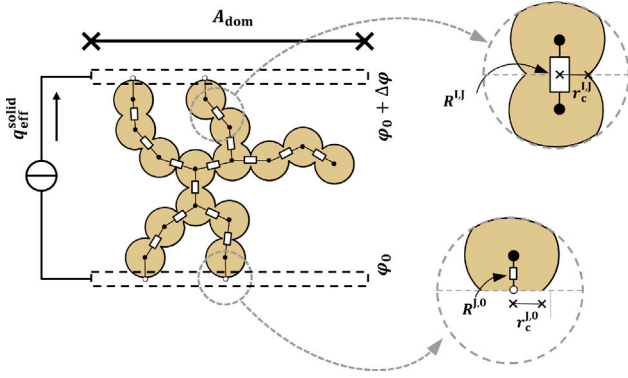


Fig. 2. Resistor network used for the calculation of the effective conductivity of the SE phase using the Resistor Network (RN) method. The SE phase is modeled as a percolating network of resistors $R^{I,J}$ between contacting particles and $R^{I,0}$ between particles and the top/bottom virtual electrodes. Each resistance is determined by its corresponding contact radius $r_c^{I,J}$ or $r_c^{J,0}$. A fixed potential difference is applied across the domain, with $\varphi_0 = 0$ on the bottom boundary and $\varphi_1 = \varphi_0 + \Delta\varphi$ on the top. The resulting steady-state ionic current q_{eff} is used along with the domain height L and cross-sectional area A_{dom} to compute the effective ionic conductivity of the SE microstructure.

with L as the domain height, A_{dom} as the cross-sectional area, and $\Delta\varphi$ as the applied potential difference. An intrinsic ionic conductivity of $\sigma = 1$ S/m was used, such that the computed $\sigma_{\text{eff,ion}}$ represents the relative effective conductivity of the SE phase. This normalization approach enables direct comparison of structure-dependent trends across simulations. Scaling the intrinsic conductivity to the experimentally measured bulk SE value would uniformly scale the resulting $\sigma_{\text{eff,ion}}$ values without affecting the relative trends or microstructural correlations. Incorporating a spatial distribution of local σ values to represent material heterogeneity (e.g., surface contamination or partial densification) could refine quantitative agreement with experiment, particularly under CAM-rich conditions, but does not alter the physical interpretation of the RNM framework.

A schematic overview of the overall FEM–RNM workflow is shown in Fig. 1(b), which outlines both the FEM-based continuum-scale analysis and the parallel RNM-based discrete transport model applied to the same DEM-derived microstructures.

3.4. Solid electrolyte network analysis: SE-utilization, SE-SE contact resistance, SE-SE coordination number

To quantify the utilization of the SE phase in composite cathodes, the SE-utilization θ_{SE} was evaluated based on the percolation connectivity of SE particles to the electrode boundaries. Following the methodology outlined in our previous work [33], only those SE particles that formed a continuous percolating path between the bottom and top surfaces of the simulation domain were considered “connected”. The SE-utilization was then defined as the volume fraction of connected SE particles relative to the total SE volume in the system

$$\theta_{\text{SE}} = \frac{V_{\text{SE}}^{\text{connected}}}{V_{\text{SE}}^{\text{total}}} \quad (16)$$

where $V_{\text{SE}}^{\text{connected}}$ is the total volume of percolating SE particles and $V_{\text{SE}}^{\text{total}}$ is the total SE volume in the microstructure.

The average contact resistance between SE particles, denoted as $\bar{R}_{\text{SE-SE}}$, was calculated to assess the typical contact resistance encountered within the percolating SE network. This quantity was derived directly from the RNM, where resistive connections were assigned between physically contacting SE particles. Assuming all particles are composed of the same SE material with uniform bulk conductivity σ ,

Table 4

Summary of particle size distribution parameters for SE and CAM powders.

Material	D_{10} [μm]	D_{50} [μm]	$\bar{D}$ [μm]	D_{90} [μm]
SE	0.71	1.49	2.00	3.80
CAM	3.90	5.50	6.00	7.62

the contact resistance between any two connected particles (particle 1 and particle 2) is given by

$$R^{1,2} = \frac{1}{2\sigma r_c^{1,2}} \quad (17)$$

where $r_c^{1,2}$ is the contact radius at the interface between the two contacting particles. The average SE–SE contact resistance was then computed over all such connected particle pairs in the percolating SE network as

$$\bar{R}_{\text{SE-SE}} = \frac{1}{N_{\text{SE-SE}}} \sum_{k=1}^{N_{\text{SE-SE}}} R_k^{1,2} \quad (18)$$

where $N_{\text{SE-SE}}$ is the total number of SE–SE contacts in the network and $R_k^{1,2}$ denotes the individual resistances between each contacting pair.

The coordination number quantifies the average number of direct SE–SE contacts per SE particle in the domain. Specifically, the SE–SE coordination number $Z_{\text{SE-SE}}$ was calculated by counting the number of neighboring SE particles in contact with each SE particle and averaging over all SE particles

$$Z_{\text{SE-SE}} = \frac{1}{N_{\text{SE}}} \sum_{i=1}^{N_{\text{SE}}} z_i \quad (19)$$

where N_{SE} is the number of SE particles in the domain, and z_i is the number of SE–SE contacts for the i th SE particle.

4. Results and discussion

4.1. Material properties

The morphological features of the SE and CAM powders were first examined using SEM, as shown in Fig. 3(a) and (b). The SE is crystalline, while the CAM is polycrystalline and exhibits roughly spherical secondary particles composed of aggregated nanoscale primaries. Detailed XRD spectra confirming the phase composition of the SE and CAM powders are provided in the Supplementary Information (Figure S1).

The PSDs of both materials, measured using laser diffraction (Fig. 3) (c), reveal monomodal distributions. The mean particle diameter of the SE was found to be $\bar{D}_{\text{SE}} = 2.0 \mu\text{m}$, while for the CAM, the corresponding value was $\bar{D}_{\text{CAM}} = 6.0 \mu\text{m}$. The full PSD statistics are summarized in Table 4.

To assess the bulk ionic conductivity of the SE, EIS was performed on a pellet compacted at 400 MPa. The EIS spectrum, shown in Fig. 3(d), yields a bulk ionic conductivity of $\sigma_{\text{SE}} = 1.02$ mS/cm at room temperature.

4.2. Experimental effective ionic conductivity

To quantify the impact of cathode composition on transport properties, full-blocking EIS measurements were conducted for cathodes with varying f_{CAM} contents of 70, 75, and 80 wt.%. The impedance spectra and corresponding TLM fits are shown in Fig. 4. As the CAM fraction increases, the total resistance associated with ionic transport through the SE network increases, resulting in a decrease in the extracted $\sigma_{\text{eff,ion}}$. The average values of $\sigma_{\text{eff,ion}}$ were 0.137 mS/cm, 0.101 mS/cm, and 0.065 mS/cm for $f_{\text{CAM}} = 70, 75$, and 80 wt.%, respectively.

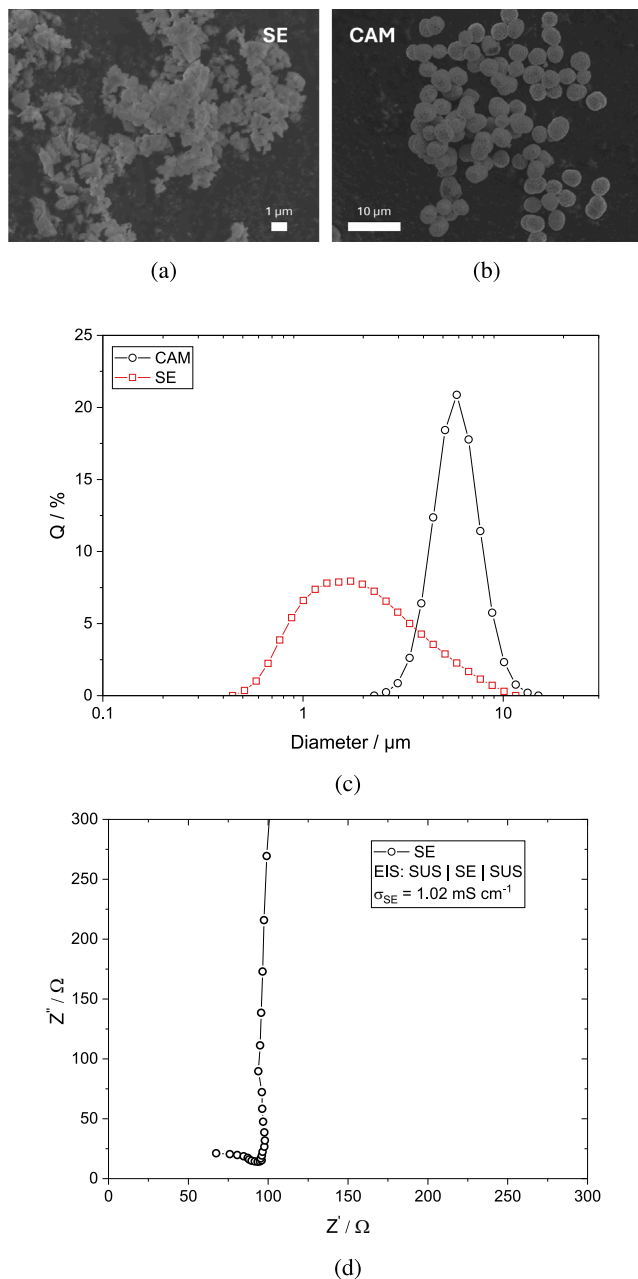


Fig. 3. Characterization of the solid electrolyte (SE) and cathode active material (CAM): (a) SEM image of SE particles showing morphology and size uniformity; (b) SEM image of CAM particles, (c) particle size distribution of SE and CAM powders; (d) EIS measurement of SE showing bulk ionic conductivity.

This direct relationship between SE content and ionic conductivity is consistent with earlier findings [12,47]. Ohno et al. [47] demonstrated that even with highly conductive SEs, a reduction in the SE fraction within the composite leads to a significant drop in effective ionic conductivity, attributed to disrupted ion percolation and geometrical constraints. Similarly, Minnmann et al. [12] reported that composites with increasing CAM content exhibit diminished ionic conductivity despite maintaining high SE bulk conductivity, highlighting the role of microstructure-induced tortuosity. These results confirm that the effective ionic conductivity is governed not only by material properties but also by phase connectivity within the cathode.

The extracted values and their standard deviations are summarized in Fig. 6(c), alongside simulation-based results discussed in subsequent

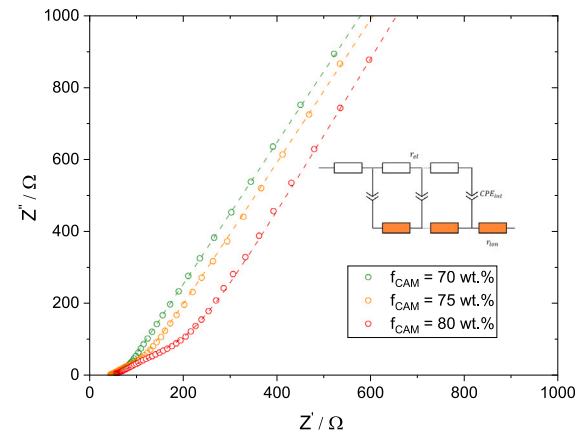


Fig. 4. Electrochemical impedance spectra (hollow dots) and corresponding transmission line model (TLM) fits (dashed lines) for full-blocking cells assembled with composite cathodes containing $f_{\text{CAM}} = 70, 75,$ and 80 wt.%. The spectra reflect the ionic transport behavior through the SE network within the cathode microstructure. An equivalent TLM circuit was used to extract the effective ionic conductivity $\sigma_{\text{eff,ion}}$ for each composition.

Table 5

Average simulation times at box factor 35 for FEM and RNM methods across different cathode compositions.

Method	f_{CAM} [wt.%]		
	70	75	80
FEM Simulation Time [s]	3352	3131	2551
RNM Simulation Time [s]	105	58	26
FEM/RNM Factor	32	54	98

sections. These experimental results will serve as a reference for validating the numerical predictions from FEM and RNM approaches later in this study.

4.3. Convergence analysis

To ensure statistical robustness and computational efficiency, a convergence analysis was conducted on both FEM and RNM models using cubic representative volume elements (RVEs) extracted from the center of the full DEM microstructures. The RVEs were defined by a unitless box factor, corresponding to cube dimensions ranging from $10 \times 10 \times 10 \mu\text{m}^3$ to $45 \times 45 \times 45 \mu\text{m}^3$. For each $f_{\text{CAM}} = 70, 75,$ and 80 wt.%, ten independently sampled RVEs were simulated per box factor. The effective ionic conductivity results for each case are presented in Fig. 5, where the top row corresponds to FEM and the bottom to RNM. The distributions reveal that beyond a box factor of approximately 35, both the mean and spread of $\sigma_{\text{eff,ion}}$ converge for all three CAM compositions and both modeling approaches. As such, a box factor of 35 was selected for subsequent simulations to balance representativity and computational cost.

To understand the influence of RVE size on microstructure and solver performance, the average number of SE particles within each RVE and the average simulation time were evaluated. As shown in Fig. 6(a), the number of SE particles increases monotonically with box factor due to volumetric scaling of the cubic RVE. This trend is consistent across all CAM fractions. Fig. 6(b) highlights the computational cost associated with each modeling approach. FEM simulations show significantly higher time requirements than RNM, especially at lower f_{CAM} where SE content is higher. For example, at box factor 35, the average simulation time for FEM was 3352 s, 3131 s, and 2551 s for $f_{\text{CAM}} = 70, 75,$ and 80 wt.%, respectively, while the RNM only required 105 s, 58 s, and 26 s. These values are summarized in Table 5.

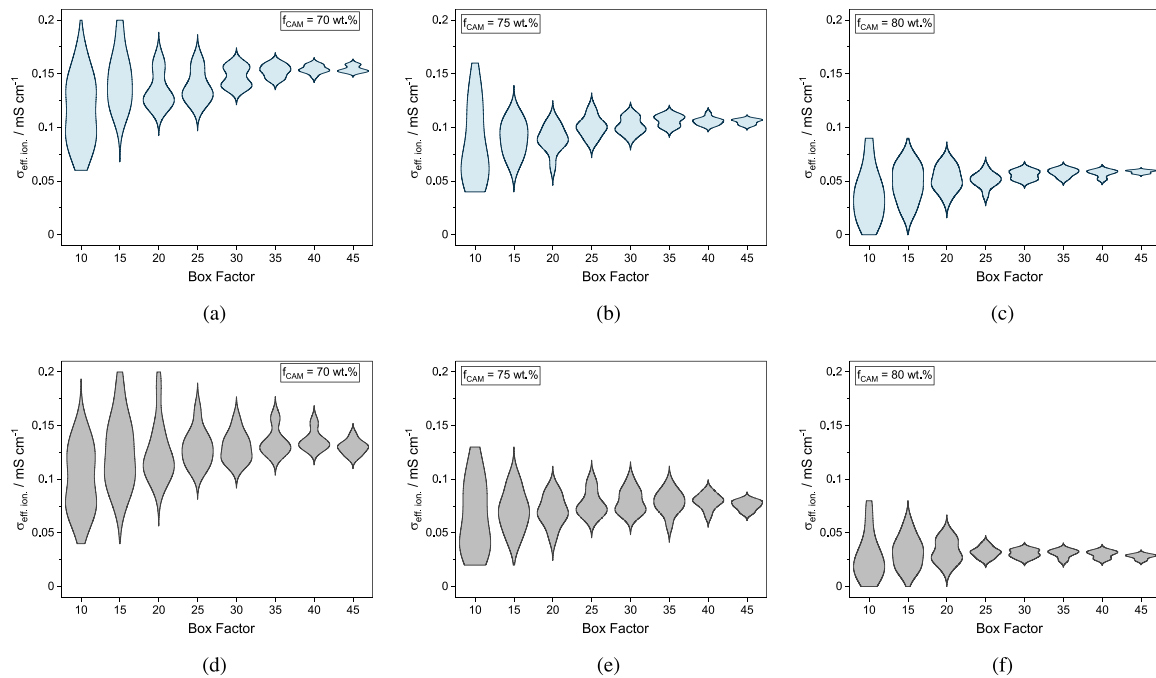


Fig. 5. Convergence behavior of $\sigma_{\text{eff,ion}}$ as a function of box factor for (a–c) Finite Element Modeling and (d–f) Resistor Network Modeling at 70, 75, and 80 wt.% f_{CAM} , respectively. Each distribution represents 10 independently extracted RVEs.

Beyond raw simulation time, RNM holds additional computational advantages. Unlike FEM, which requires preprocessing steps such as SE phase preparation and mesh generation (not included in the timing analysis), RNM operates directly on particle-level data. This distinction is also illustrated in the workflow schematic (Fig. 1) (b), where the more streamlined RNM path bypasses time-intensive volumetric meshing steps required for FEM. This drastically reduces total workflow time and simplifies automation for high-throughput simulations. Given its efficiency, RNM provides a scalable framework for assessing microstructure-transport relationships across large design spaces.

4.4. Model validation

To evaluate the predictive accuracy of the numerical models, the effective ionic conductivity values computed from FEM and RNM were compared against experimental measurements for cathodes with $f_{\text{CAM}} = 70, 75$, and 80 wt.%. The results are summarized in Fig. 6(c), which includes the mean values and standard deviations across all three methods. Both FEM and RNM models reproduce the expected monotonic decrease in $\sigma_{\text{eff,ion}}$ with increasing CAM content. At $f_{\text{CAM}} = 70$ wt.%, all three values are in close agreement. At 75 wt.%, FEM shows excellent alignment with the experimental measurement, while RNM underpredicts slightly. At the highest CAM fraction (80 wt.%), RNM shows the largest deviation, underestimating the effective ionic conductivity, whereas FEM remains closer to the measured value.

Overall, RNM tends to slightly underestimate $\sigma_{\text{eff,ion}}$ across all compositions, particularly at high CAM loading. This discrepancy does not arise from missing percolating pathways, since both FEM and RNM operate on the same DEM-derived SE network topology. Instead, it stems from the simplified treatment of geometrical ionic transport in RNM, which represents conduction solely through idealized constriction-type resistances between contacting particles. In doing so, RNM neglects the continuous potential distribution and multi-contact current spreading captured by FEM within the SE phase. Furthermore, the DEM-derived particle overlap directly influences the computed contact radii and, consequently, the resistances: large overlaps at high SE content drive the local resistances toward zero, while minimal overlaps at high CAM content lead to extremely large resistances. For high SE contents, the

void fraction is also minimal, further increasing interparticle overlaps and thereby reducing local constriction resistances. This behavior reflects the composition-dependent contact quality and amplifies the inherent simplifications of RNM at the extremes of SE content. Despite these limitations, RNM reproduces the correct physical trends and remains in good agreement with both FEM and experimental results in our studied window, demonstrating its reliability as a computationally efficient surrogate for detailed field-resolved simulations. In future work, the RNM framework could be calibrated against FEM or experimental data by introducing an empirical correction factor to account for field-spreading and local constriction effects, thereby further improving quantitative agreement at composition extremes.

4.5. Solid electrolyte particle size variation

The influence of SE particle size on $\sigma_{\text{eff,ion}}$ was investigated using FEM and RNM calculations on DEM-generated microstructures. Motivated by prior studies highlighting the critical role of the SE-to-CAM particle size ratio in determining contact quality, percolation pathways, and overall ionic transport in composite cathodes [13,19,33,57], we systematically varied the SE mean diameter from 1 μm to 8 μm under a constant compaction pressure of 400 MPa. Cathode compositions with f_{CAM} between 70 and 80 wt.% were analyzed to assess the combined effects of particle size and composition on the SE transport network.

As shown in Fig. 7 (a and b), both FEM and RNM results exhibit a clear and consistent trend: $\sigma_{\text{eff,ion}}$ increases with decreasing f_{CAM} and decreasing $\bar{D}_{\text{SE}}$, indicating enhanced ionic percolation with finer SE particles. This improvement can be attributed to the higher surface area and packing density of smaller SE particles, which collectively enhance the number and quality of interparticle contacts and facilitate better percolating pathways. While RNM slightly underestimates the conductivity compared to FEM across the range, the trends remain well-aligned, reinforcing the reliability of both models. The increasing deviation at finer sizes may stem from RNM's constriction-based simplifications, which overlook complex field-level effects captured in FEM. Nevertheless, RNM successfully captures the dominant dependencies and provides rapid estimations at a fraction of the computational

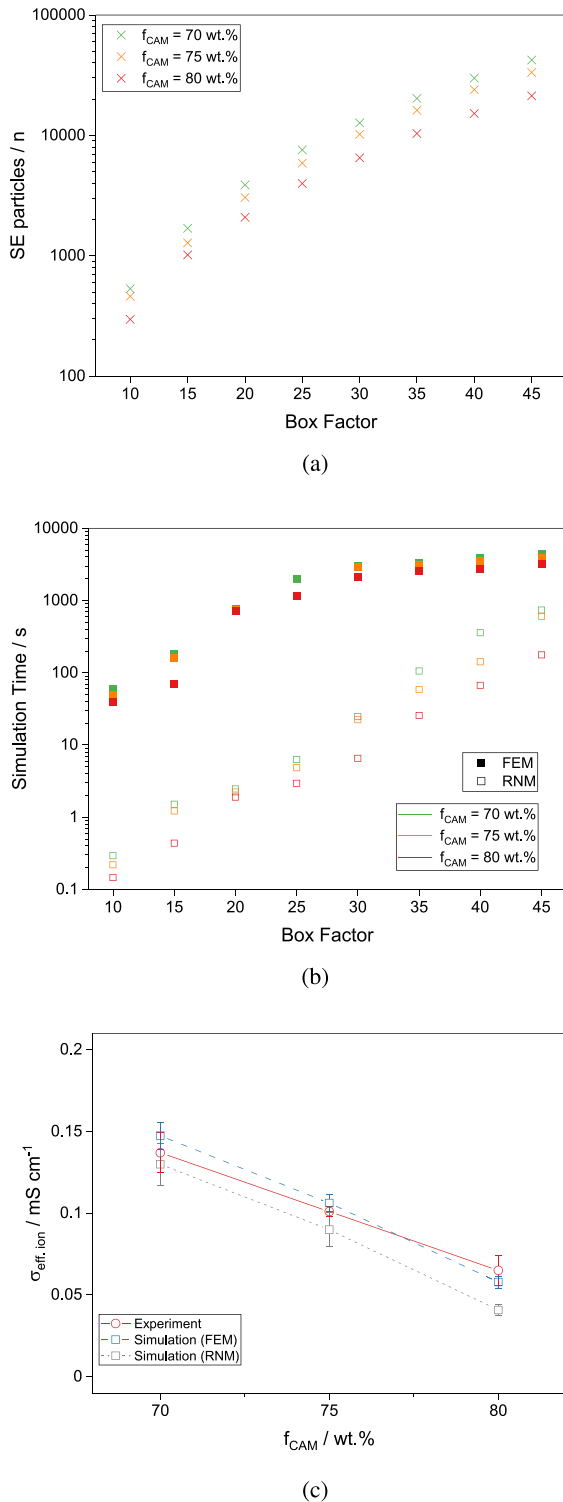


Fig. 6. (a) Average number of SE particles within the RVE as a function of box factor, showing a monotonic increase due to volumetric scaling. (b) Average simulation time required for FEM and RNM solvers as a function of box factor for different CAM contents. RNM exhibits significantly shorter simulation times than FEM across all cases. (c) Comparison of effective ionic conductivity $\sigma_{eff,ion}$ extracted from experimental full-blocking EIS measurements, Finite Element Modeling (FEM), and Resistor Network Modeling (RNM) for composite cathodes with $f_{CAM} = 70, 75$, and 80 wt.%.

cost. Overall, the results highlight the critical influence of SE size engineering in optimizing ionic transport in composite cathodes.

To further interpret the conductivity trends, three network descriptors were evaluated: θ_{SE} , Z_{SE-SE} , and $\bar{R}_{SE-SE}$, which quantify SE utilization, coordination number, and average SE-SE contact resistance, respectively.

As shown in Fig. 7 (c–e), θ_{SE} increases with decreasing f_{CAM} and smaller $\bar{D}_{SE}$, reflecting a larger percolating SE fraction in microstructures with finer SE particles and greater SE content. Similarly, Z_{SE-SE} follows the same trend, confirming that smaller particles enhance local connectivity through a higher number of particle–particle contacts.

The average SE-SE contact resistance $\bar{R}_{SE-SE}$ decreases significantly with increasing $\bar{D}_{SE}$, which might initially appear counterintuitive. However, this can be attributed to the fact that while small SE particles create many contacts (high Z_{SE-SE}), the contact areas (and thus the contact radii r_c) are small, leading to high individual resistances. This is explicitly evident from the inverse proportionality of $R^{i,j} \propto 1/r_c^{i,j}$. Nonetheless, even when $\bar{R}_{SE-SE}$ is elevated at small $\bar{D}_{SE}$, the concurrent increase in θ_{SE} and Z_{SE-SE} leads to a more robust and highly interconnected SE percolation network. The larger number of parallel transport paths enabled by these additional contacts offsets the impact of higher individual resistances, ultimately resulting in improved effective ionic conductivity $\sigma_{eff,ion}$. Moreover, at high f_{CAM} , the SE is more mechanically isolated due to the surrounding stiffer CAM particles. As discussed by Ohashi et al. [18], the presence of stiffer CAM redistributes mechanical stress away from the SE phase, reducing densification and inhibiting the formation of intimate SE-SE contacts. This mechanical effect further elevates $\bar{R}_{SE-SE}$ in CAM-rich composites.

Our findings confirm that reducing the SE particle size enhances the formation of percolating ionic pathways, particularly at high f_{CAM} . Smaller SE particles are more effective at filling voids between CAM particles and forming a larger number of contacts, which improves overall network connectivity. Additionally, maintaining a high CAM-to-SE particle size ratio promotes microstructural interpenetration, allowing for continuous ionic transport even at low SE volume fractions. These structural improvements directly translate into higher effective ionic conductivity $\sigma_{eff,ion}$, which we define as the macroscopic transport property of the SE network derived from local particle connectivity and interparticle contact resistances. Although the use of fine SE particles to enhance ionic transport has been established in prior studies [13,19,57], our results provide quantitative microstructural validation across a range of CAM loadings. In particular, we demonstrate that the metrics θ_{SE} , Z_{SE-SE} , and $\bar{R}_{SE-SE}$ serve as effective indicators of percolation quality and contact efficiency, which are critical for maintaining conductivity in CAM-rich electrodes. Higher θ_{SE} and Z_{SE-SE} , along with lower $\bar{R}_{SE-SE}$, correspond to more continuous and efficient transport paths, thereby increasing $\sigma_{eff,ion}$. These network-level descriptors can support the targeted tuning of SE particle size distributions to compensate for reduced SE content at high CAM loading. Careful particle engineering guided by these insights can facilitate the design of cathodes that achieve both high energy density and effective ion transport.

4.6. Pressure dependence

The effect of compaction pressure on the SE transport network was systematically investigated using RNM applied to DEM-generated microstructures. The simulations were based on measured particle size distributions (Fig. 3) (c) and explored cathode compositions with $f_{CAM} = 70, 75$, and 80 wt.% under pressures ranging from 100 to 400 MPa. This analysis was conducted to evaluate how compaction pressure influences the SE transport network and to assess whether DEM and RNM are capable of capturing such effects—an important consideration given prior experimental findings by Yamamoto et al. [14], who demonstrated that fabrication pressure significantly affects composite cathode structure and performance.

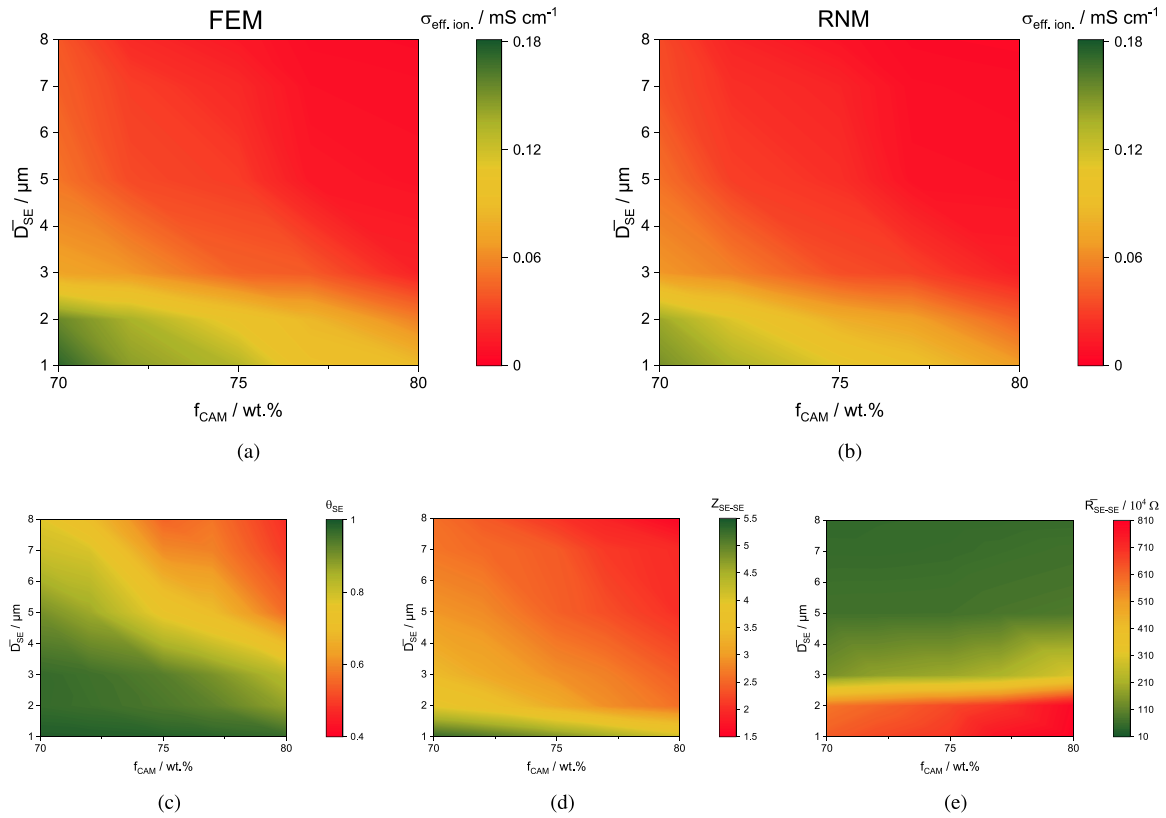


Fig. 7. Effective ionic conductivity $\sigma_{\text{eff,ion}}$ as a function of mean SE particle size $\bar{D}_{\text{SE}}$ and f_{CAM} computed using (a) FEM and (b) RNM. SE network descriptors as a function of $\bar{D}_{\text{SE}}$ and f_{CAM} : (c) SE utilization θ_{SE} , (d) SE-SE coordination number $Z_{\text{SE-SE}}$, and (e) average SE-SE contact resistance $\bar{R}_{\text{SE-SE}}$.

Fig. 8 summarizes the results from the RNM calculations. As shown in Fig. 8(a), $\sigma_{\text{eff,ion}}$ increases monotonically with pressure across all compositions. This trend is attributed to enhanced particle packing and improved SE-SE contact formation, leading to more effective percolation. From 100 to 400 MPa, the absolute and relative improvements are +0.067 mS/cm (98%) for 70 wt.%, +0.044 mS/cm (126%) for 75 wt.%, and +0.023 mS/cm (291%) for 80 wt.%. Although the relative gain is largest at 80 wt.% due to a poorly connected initial network, the highest absolute values are still reached at 70 wt.%, highlighting the importance of sufficient SE content. These observations are in agreement with Ohashi et al. [18], who reported that pressure-enhanced transport benefits are more pronounced at low active material content.

As shown in Fig. 8(b), θ_{SE} also increases with pressure, indicating a greater fraction of SE particles become percolatively active. The gains are especially prominent in the 80 wt.% case, where connectivity is initially poor. In contrast, 70 wt.% systems exhibit a high baseline θ_{SE} already above 0.9, with comparatively smaller increases.

In Fig. 8(c), $\bar{R}_{\text{SE-SE}}$ is observed to decrease with pressure, reflecting improved interparticle contact quality. The greatest reductions occur for 70 wt.%, where more SE-SE contacts exist. Higher values of $\bar{R}_{\text{SE-SE}}$ persist at 80 wt.% due to fewer available SE contacts. This observation is supported by Ohashi et al. [18], who explained that stiffer CAM particles can limit pressure transmission into the SE phase, reducing stress-induced contact formation.

The coordination number $Z_{\text{SE-SE}}$ (Fig. 8) (d) increases with pressure across all cases, consistent with densification-driven contact generation. While 70 wt.% CAM achieves the highest absolute $Z_{\text{SE-SE}}$, the relative increase is largest at 80 wt.%, where the SE phase starts from a sparse, poorly connected structure.

These results underscore the complex interdependence between compaction pressure, cathode microstructure, and ion transport performance. While all metrics improve with pressure, the degree of enhancement depends strongly on the available SE volume fraction and

the initial microstructural state. At lower f_{CAM} , the SE network is more extensive, enabling greater absolute gains in connectivity, contact quality, and ultimately conductivity. In contrast, high f_{CAM} systems benefit more in relative terms, as pressure compensates for initially poor SE phase connectivity. Moreover, the diminishing returns at higher pressures suggest that a saturation-like behavior may emerge beyond a critical densification threshold. Together, these findings emphasize that optimal design of solid-state cathodes requires balancing composition, particle size, and fabrication pressure. The combination of DEM and RNM modeling provides a powerful, mechanistically grounded approach to dissect and predict these interrelated effects on SE phase performance.

From a practical perspective, these results highlight that compaction pressure can be strategically leveraged to mitigate transport limitations, particularly in CAM-rich cathodes where the SE network is initially under-connected. While increasing pressure enhances θ_{SE} , $Z_{\text{SE-SE}}$, and reduces $\bar{R}_{\text{SE-SE}}$, the improvements in $\sigma_{\text{eff,ion}}$ begin to saturate at higher pressure levels, especially in SE-rich compositions. This suggests that an optimal pressure range exists, beyond which further densification offers limited benefit. The modeling framework presented here enables this pressure-conductivity relationship to be quantified based on cathode composition and microstructure, supporting rational process optimization for solid-state cathode fabrication.

5. Conclusion

We presented a simulation framework combining DEM-based microstructures with FEM and RNM to analyze $\sigma_{\text{eff,ion}}$ in composite cathodes for ASSBs. Convergence analysis using RVE extraction confirmed that both FEM and RNM yield stable and consistent $\sigma_{\text{eff,ion}}$ predictions beyond a box factor of 35. Importantly, both models demonstrated strong agreement with experimental results, validating their reliability in capturing realistic SE network behavior.

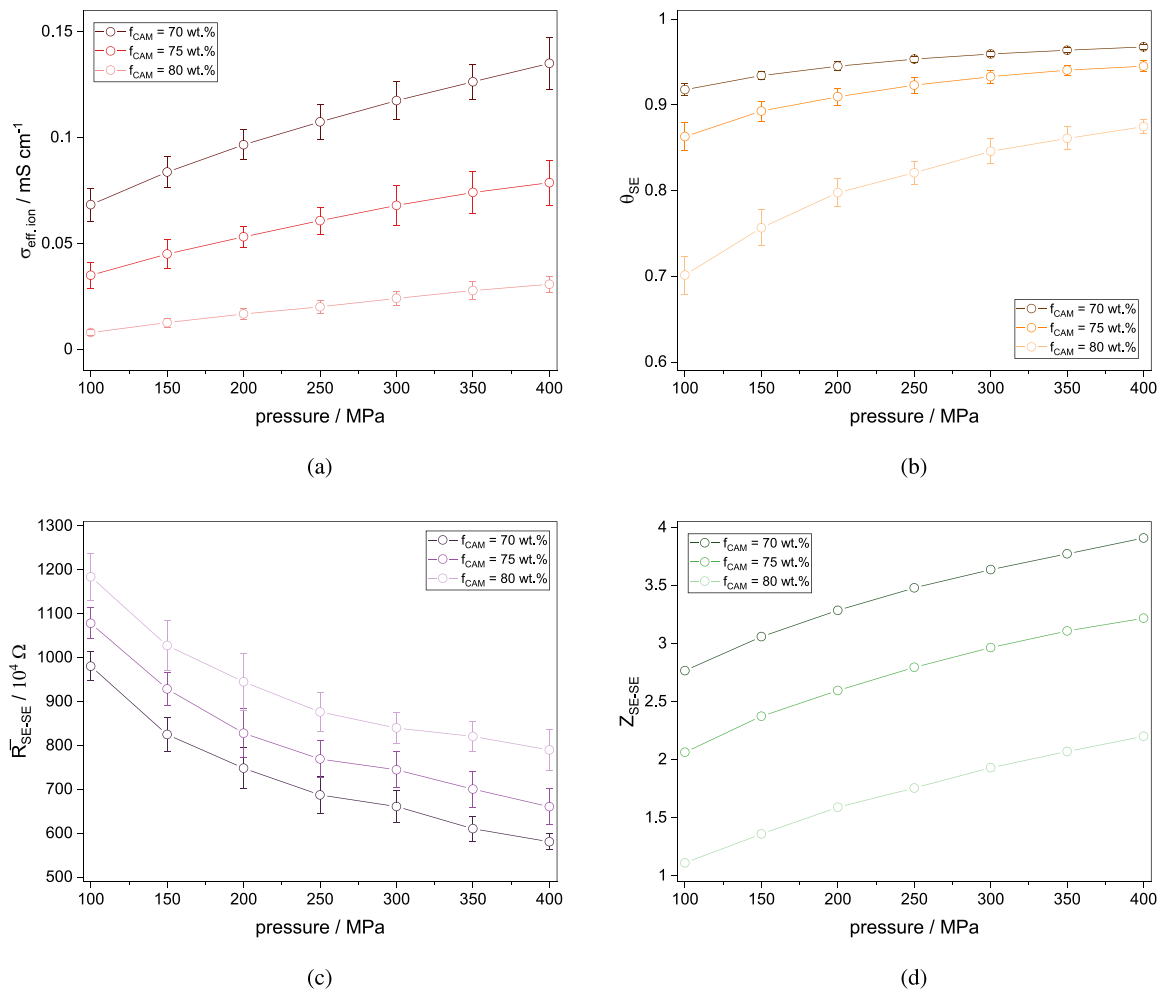


Fig. 8. Pressure-dependent properties of the SE phase evaluated using resistor network modeling (RNM): (a) effective ionic conductivity $\sigma_{eff,ion}$, (b) SE-utilization θ_{SE} , (c) average SE-SE contact resistance $\bar{R}_{SE-SE}$, and (d) SE-SE coordination number Z_{SE-SE} as a function of compaction pressure for cathodes with $f_{CAM} = 70$, 75, and 80 wt.%.

While FEM offers high-fidelity spatial resolution, RNM provided similar accuracy at drastically reduced computational cost — up to two orders of magnitude faster — without requiring mesh generation or SE phase preparation. This makes RNM especially suitable for large-scale parametric studies and high-throughput screening.

Systematic variation of $\bar{D}_{SE}$ revealed that decreasing particle size enhances $\sigma_{eff,ion}$, driven by improvements in θ_{SE} and Z_{SE-SE} . These trends highlight the critical role of the SE-to-CAM particle size ratio in microstructure design. The network analysis descriptors proved instrumental in revealing how finer $\bar{D}_{SE}$ promotes better connectivity and reduced contact resistance. While RNM slightly underestimates $\sigma_{eff,ion}$ compared to FEM, both methods show consistent trends and remain within experimental uncertainty—reinforcing the validity and complementarity of the two approaches.

A parametric study of pressure from 100–400 MPa showed consistent improvements in $\sigma_{eff,ion}$, θ_{SE} , $\bar{R}_{SE-SE}$, and Z_{SE-SE} with increasing compaction, particularly for lower f_{CAM} where SE content is higher. However, the rate of improvement diminishes at higher pressures, indicating a saturation behavior as the system approaches densification limits. The DEM and RNM framework effectively captured the pressure-induced changes in particle packing, percolation pathways, and SE contact quality—highlighting their sensitivity to realistic processing conditions.

This integrated modeling approach offers a fast and predictive path to microstructure optimization in ASSBs. Beyond its current scope, the framework can be extended to incorporate electronic transport

pathways alongside ionic conduction for a more comprehensive representation of the composite cathode. Although electronic percolation is typically ensured experimentally through the addition of a small fraction of conductive additives such as CNFs, its explicit inclusion would enable coupled ion–electron transport analysis. This can be feasibly achieved through digital post-processing integration of CNFs within the DEM-generated microstructure, allowing the formation of a secondary conductive network without significantly increasing computational cost. Furthermore, while DEM inherently assumes spherical particles for numerical efficiency and stable contact resolution, a multi-sphere DEM approach could be employed in future work to approximate faceted or irregular particle morphologies, albeit at considerably higher computational expense. Combined with interfacial contact modeling, these potential extensions would enhance the physical completeness of the DEM–FEM–RNM framework and its applicability to realistic cathode architectures.

CRediT authorship contribution statement

Ali Mohammad Bazzoun: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Javid Piruzjam:** Writing – review & editing, Methodology. **Steffen Hink:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Lukas Rubacek:** Writing – review & editing, Supervision, Funding acquisition. **Alexander Fill:** Writing – review & editing, Supervision. **Kai Peter Birke:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used DeepL Write and ChatGPT (version 3.5 by OpenAI) to partially improve the language, readability, and grammar of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2025.238682>.

Data availability

Data will be made available on request.

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